Dataset Validation

Surgical Scene Semantic Segmentation

Dataset: cholec_seg8k

Number of classes: 13

Complete-dataset distribution analysis

This report evaluates class presence, pixel-level distribution, split consistency, and qualitative segmentation annotations.

All quantitative statistics are computed using the complete samples available in each split.

1. Dataset Overview

Dataset: cholec_seg8k

Split	Number of samples
train	5360
validation	2720
test	0

Classes

```
 0  —  Background
 1  —  Abdominal Wall
 2  —  Liver
 3  —  Gastrointestinal Tract
 4  —  Fat
 5  —  Grasper
 6  —  Connective Tissue
 7  —  Blood
 8  —  Cystic Duct
 9  —  L-hook Electrocautery
10  —  Gallbladder
11  —  Hepatic Vein
12  —  Liver Ligament
```

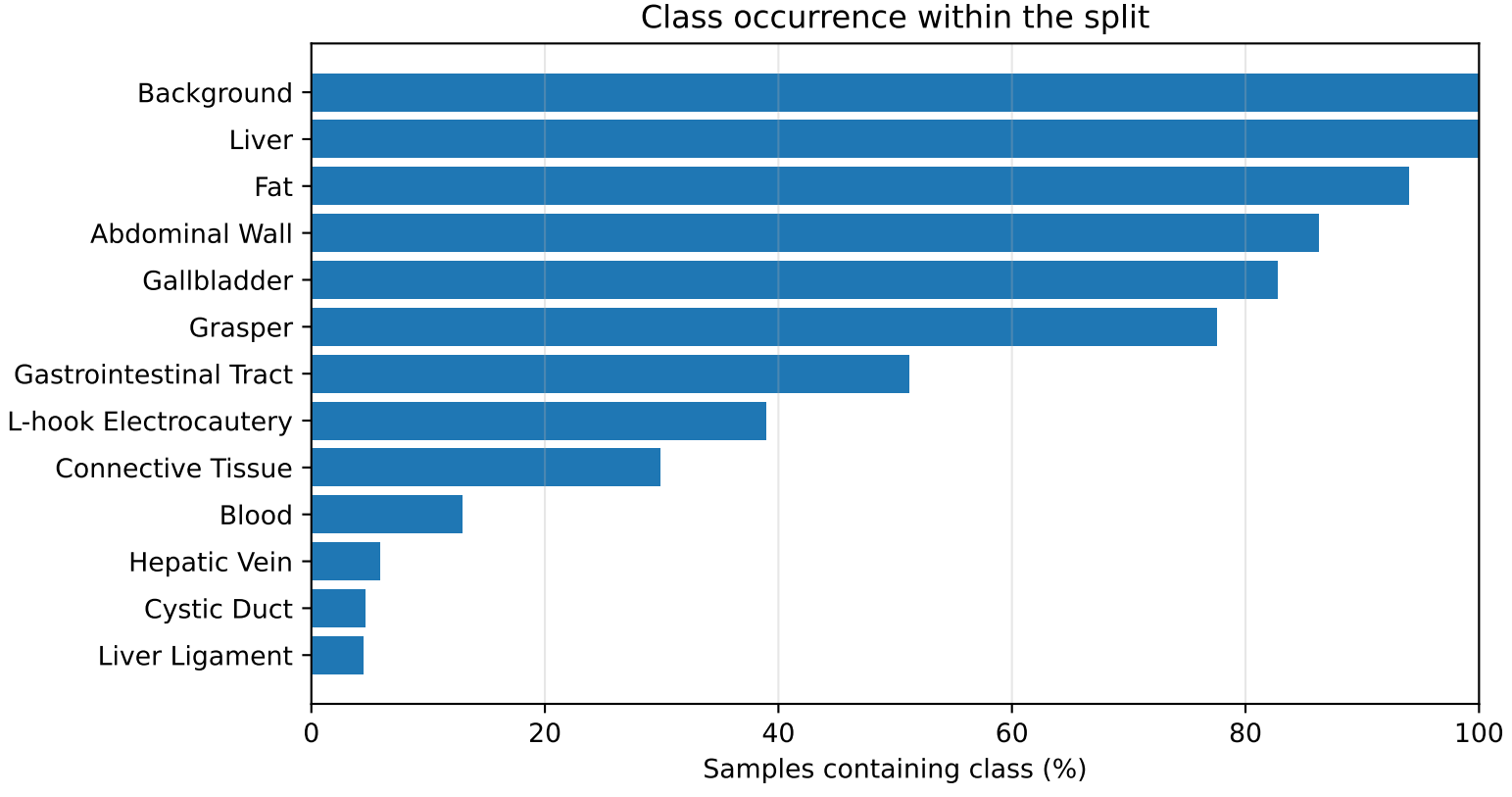
Methodology

The validation procedure evaluates every sample in each split. No sampling or maximum-sample limit is applied to the quantitative analysis.

2. Class Presence

Split: train

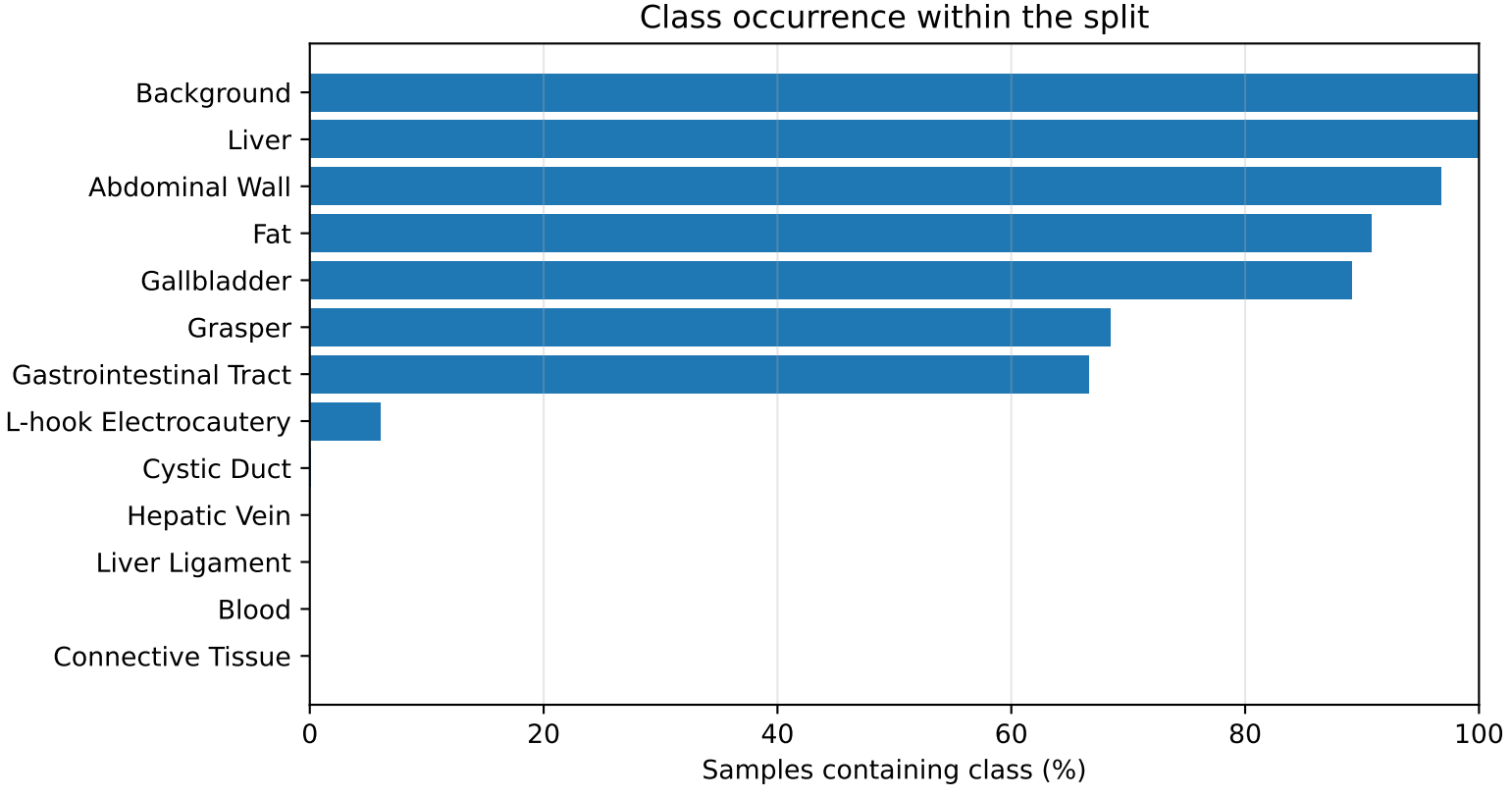
Class	Samples	Samples containing class
Background	5,360	100.00%
Abdominal Wall	4,624	86.27%
Liver	5,360	100.00%
Gastrointestinal Tract	2,746	51.23%
Fat	5,039	94.01%
Grasper	4,158	77.57%
Connective Tissue	1,600	29.85%
Blood	692	12.91%
Cystic Duct	246	4.59%
L-hook Electrocautery	2,089	38.97%
Gallbladder	4,436	82.76%
Hepatic Vein	317	5.91%
Liver Ligament	240	4.48%



2. Class Presence

Split: validation

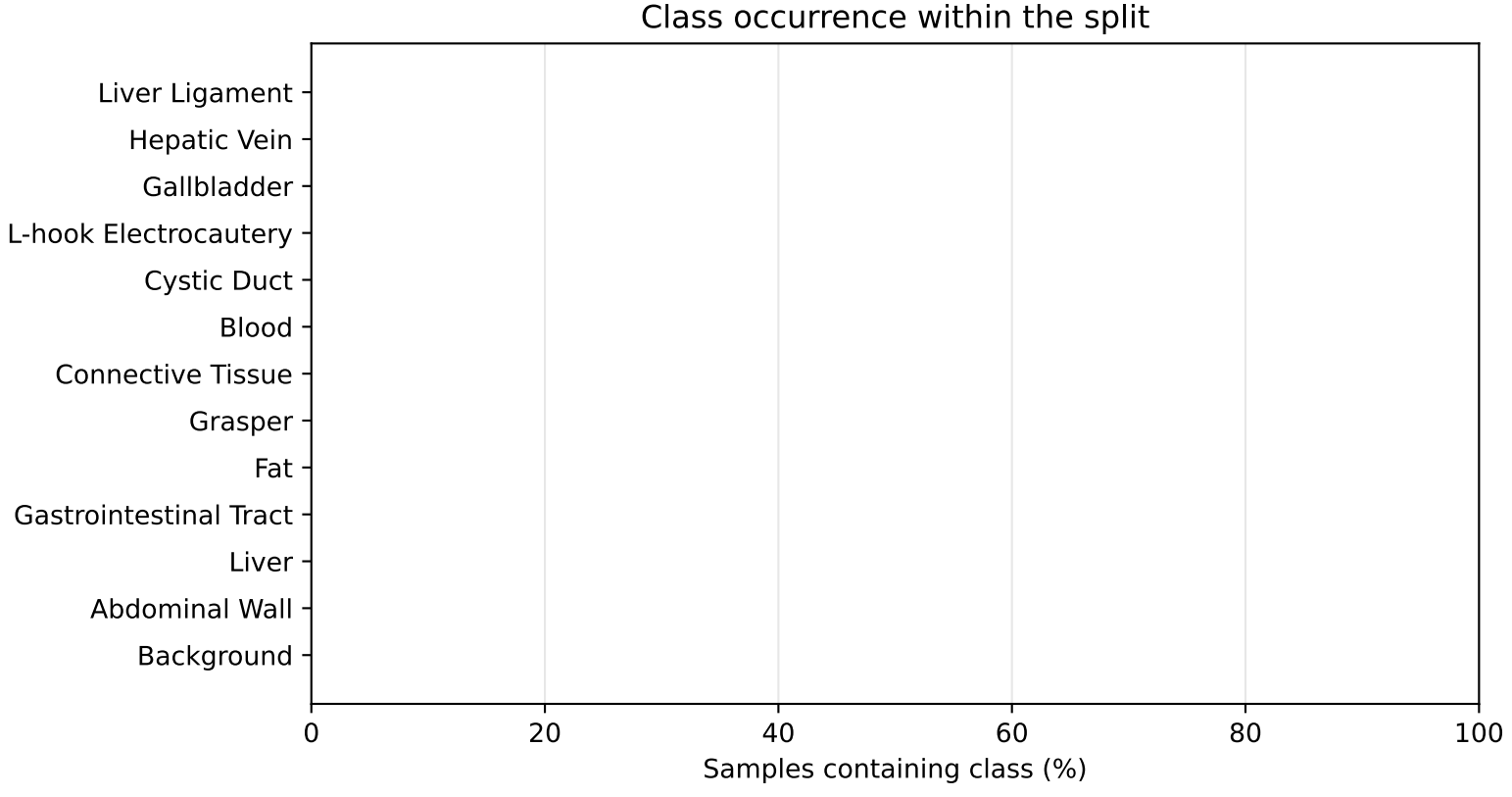
Class	Samples	Samples containing class
Background	2,720	100.00%
Abdominal Wall	2,631	96.73%
Liver	2,720	100.00%
Gastrointestinal Tract	1,812	66.62%
Fat	2,471	90.85%
Grasper	1,862	68.46%
Connective Tissue	0	0.00%
Blood	0	0.00%
Cystic Duct	2	0.07%
L-hook Electrocautery	165	6.07%
Gallbladder	2,425	89.15%
Hepatic Vein	0	0.00%
Liver Ligament	0	0.00%



2. Class Presence

Split: test

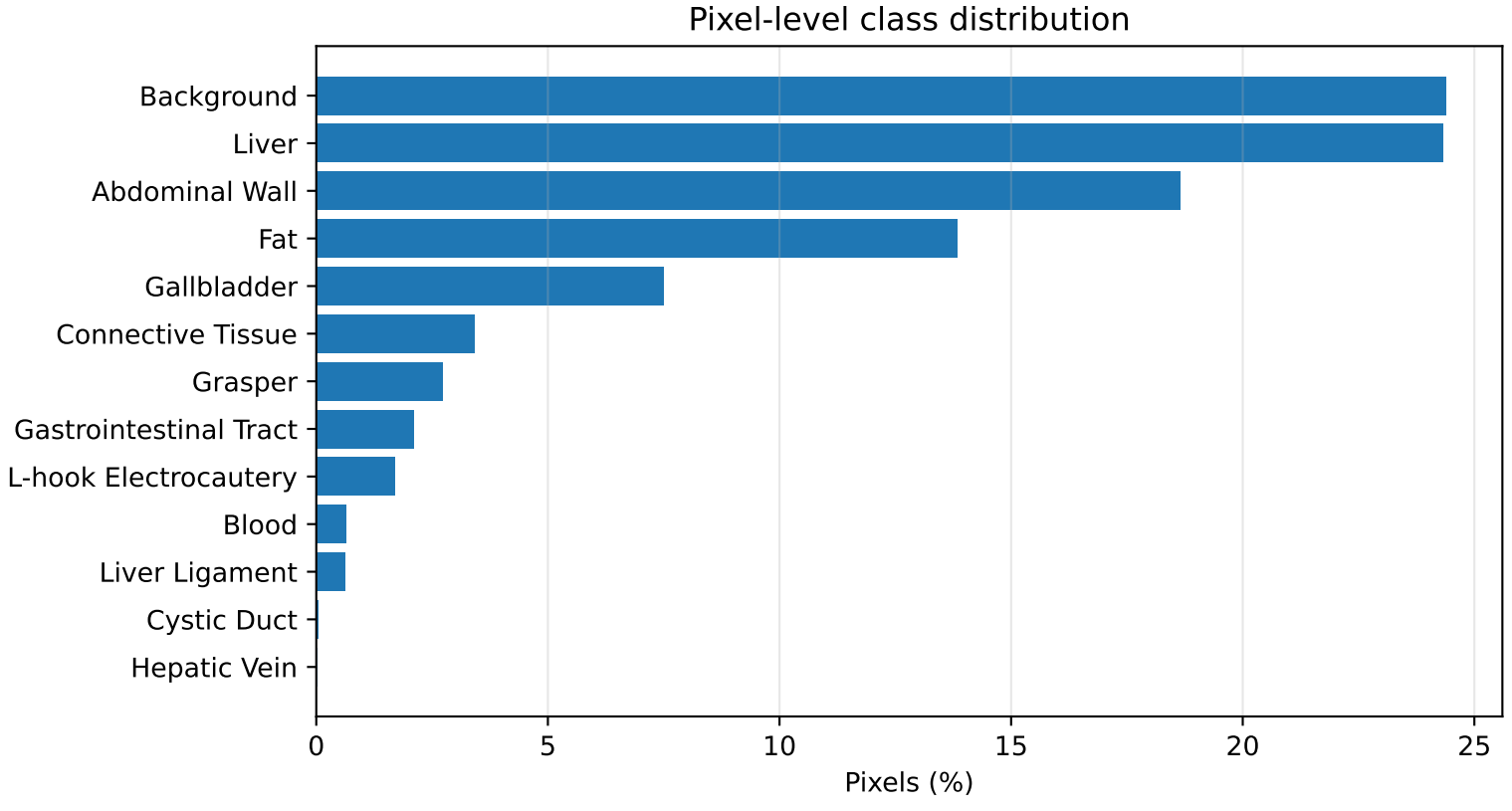
Class	Samples	Samples containing class
Background	0	0.00%
Abdominal Wall	0	0.00%
Liver	0	0.00%
Gastrointestinal Tract	0	0.00%
Fat	0	0.00%
Grasper	0	0.00%
Connective Tissue	0	0.00%
Blood	0	0.00%
Cystic Duct	0	0.00%
L-hook Electrocautery	0	0.00%
Gallbladder	0	0.00%
Hepatic Vein	0	0.00%
Liver Ligament	0	0.00%



3. Pixel Distribution

Split: train

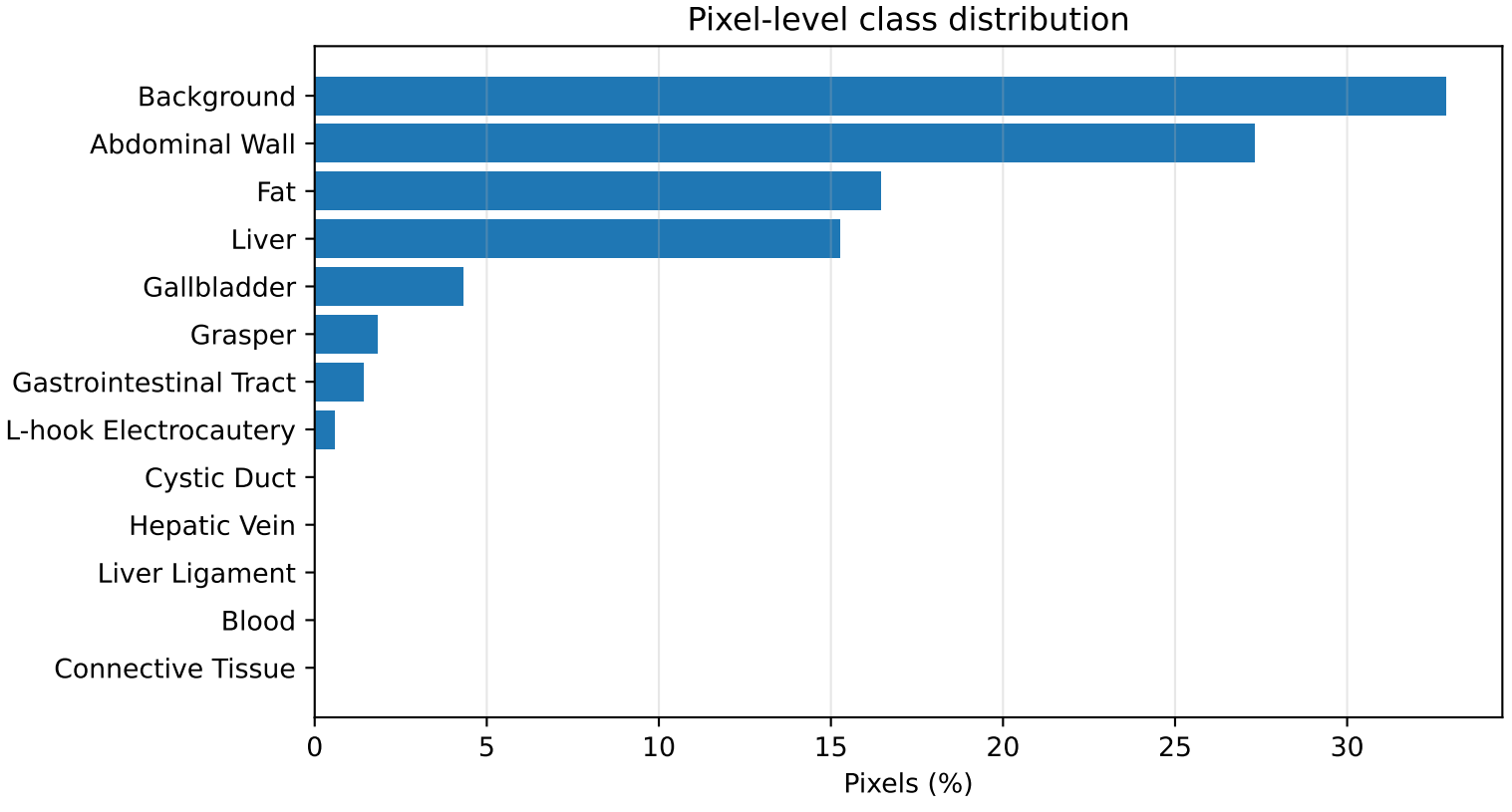
Class	Pixels	Pixel distribution
Background	535,797,674	24.3858%
Abdominal Wall	410,013,131	18.6610%
Liver	534,385,476	24.3215%
Gastrointestinal Tract	46,409,566	2.1122%
Fat	304,187,615	13.8445%
Grasper	60,016,433	2.7315%
Connective Tissue	75,080,239	3.4171%
Blood	13,991,063	0.6368%
Cystic Duct	1,155,394	0.0526%
L-hook Electrocautery	37,084,296	1.6878%
Gallbladder	164,866,729	7.5036%
Hepatic Vein	314,317	0.0143%
Liver Ligament	13,869,267	0.6312%



3. Pixel Distribution

Split: validation

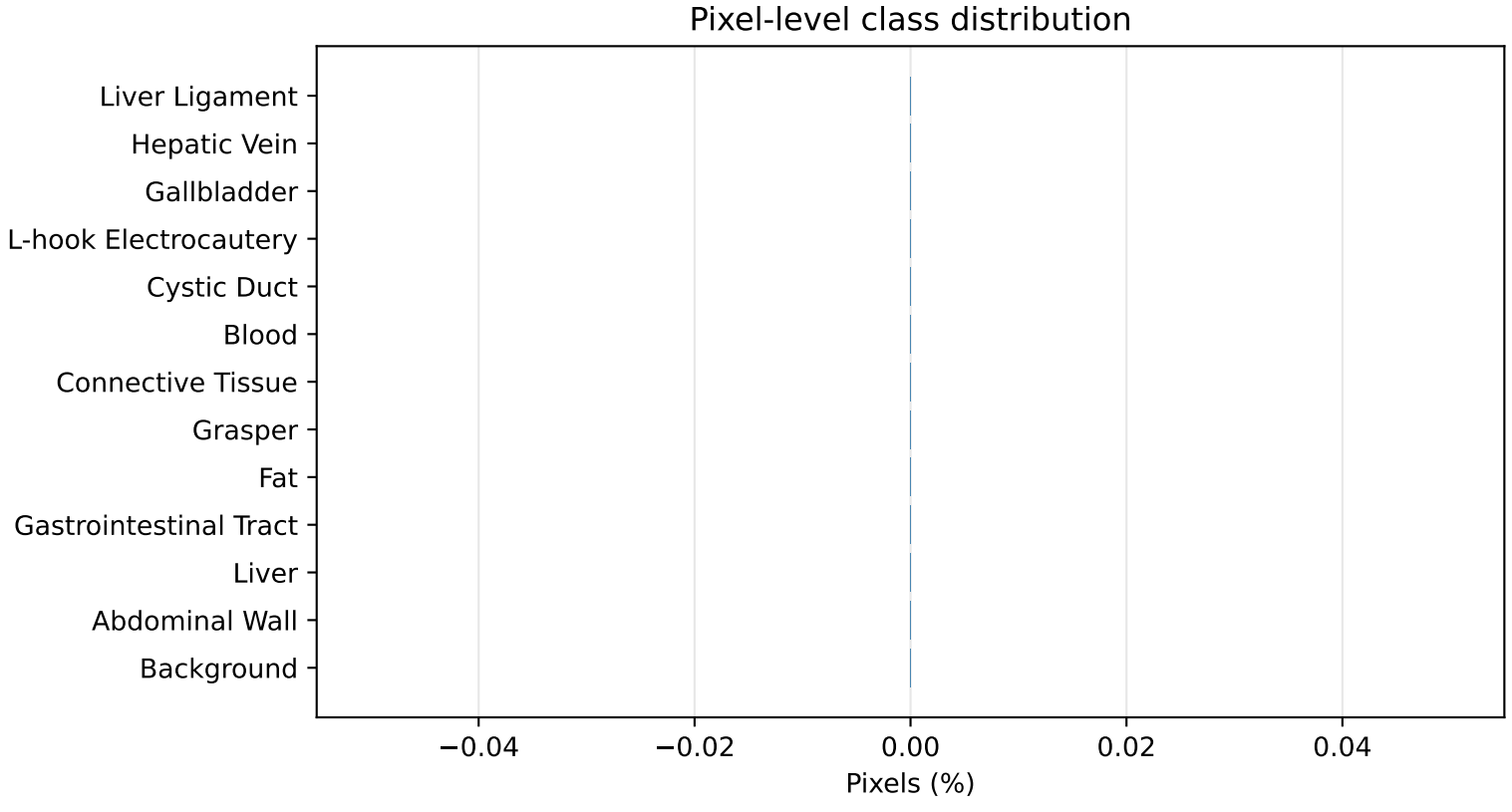
Class	Pixels	Pixel distribution
Background	366,459,488	32.8668%
Abdominal Wall	304,384,903	27.2995%
Liver	170,154,449	15.2607%
Gastrointestinal Tract	15,880,304	1.4243%
Fat	183,377,743	16.4467%
Grasper	20,198,514	1.8116%
Connective Tissue	0	0.0000%
Blood	0	0.0000%
Cystic Duct	66,817	0.0060%
L-hook Electrocautery	6,348,766	0.5694%
Gallbladder	48,111,416	4.3150%
Hepatic Vein	0	0.0000%
Liver Ligament	0	0.0000%

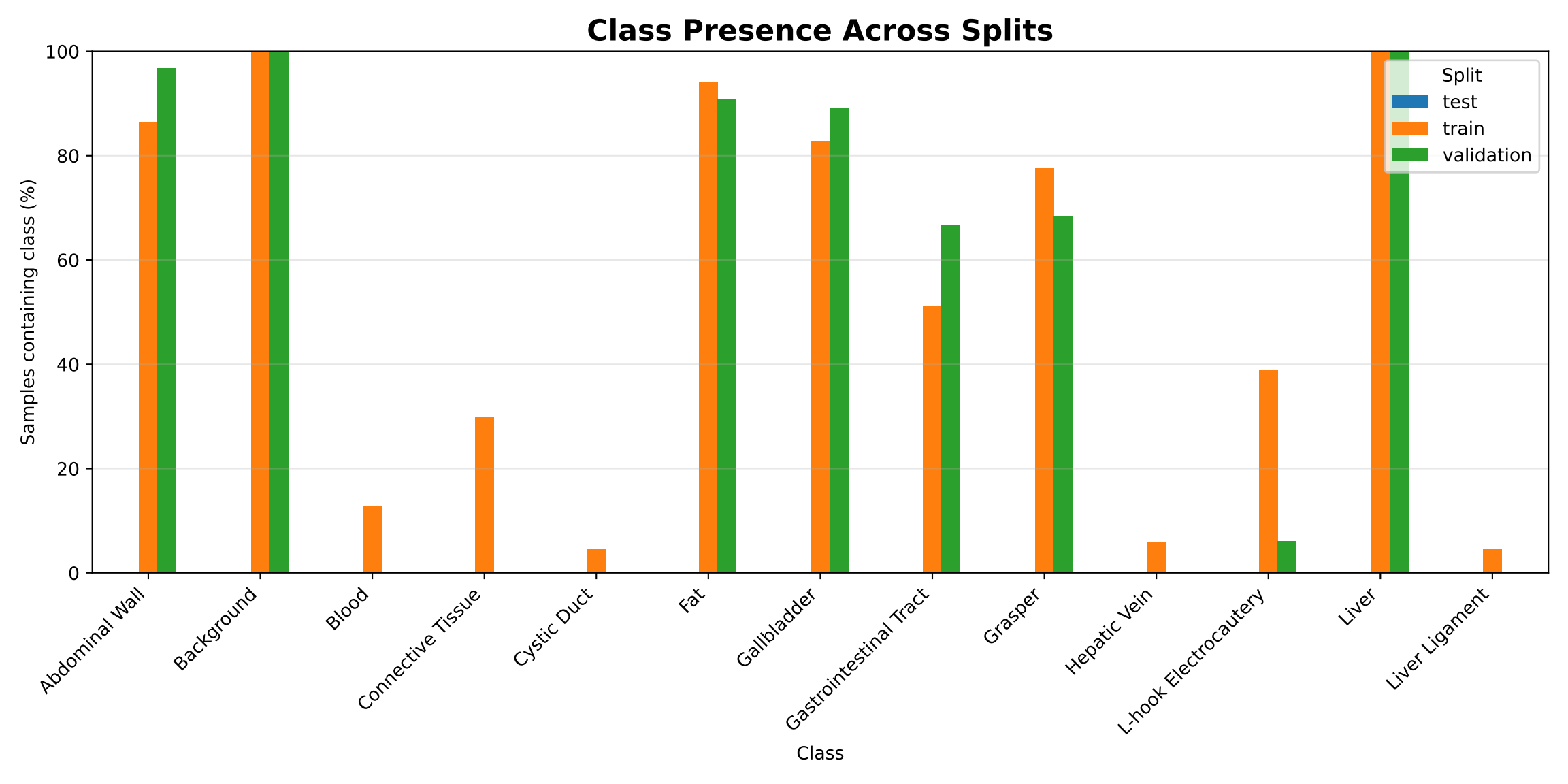


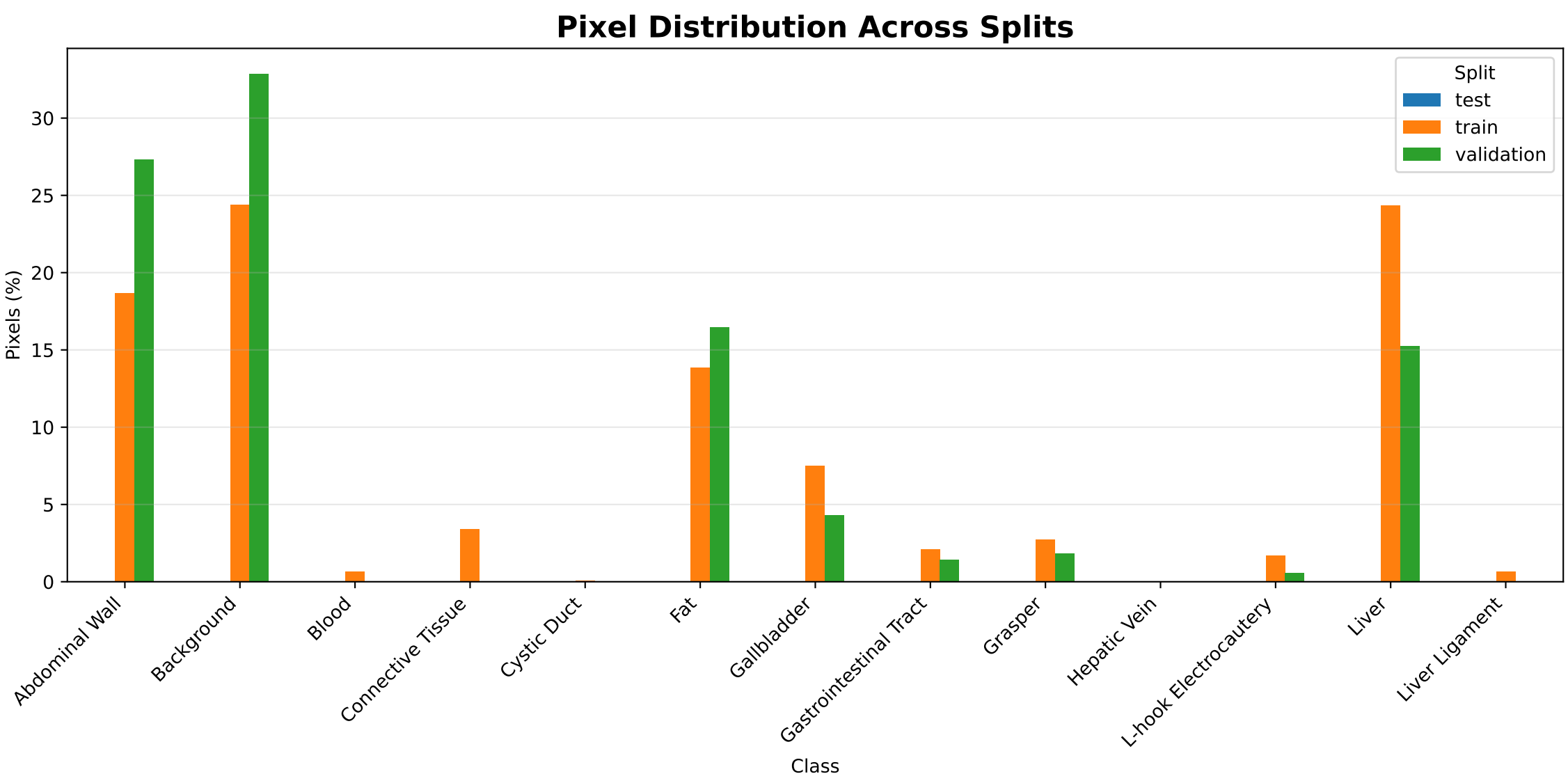
3. Pixel Distribution

Split: test

Class	Pixels	Pixel distribution
Background	0	0.0000%
Abdominal Wall	0	0.0000%
Liver	0	0.0000%
Gastrointestinal Tract	0	0.0000%
Fat	0	0.0000%
Grasper	0	0.0000%
Connective Tissue	0	0.0000%
Blood	0	0.0000%
Cystic Duct	0	0.0000%
L-hook Electrocautery	0	0.0000%
Gallbladder	0	0.0000%
Hepatic Vein	0	0.0000%
Liver Ligament	0	0.0000%

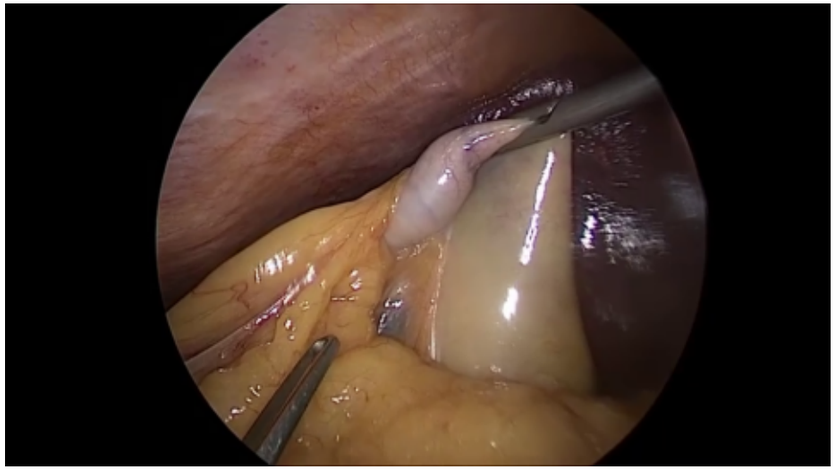



Class Presence Across Splits
Samples containing class (%)
100
80
60
40
20
0
Split
test
train
validation
Abdominal Wall
Background
Blood
Connective Tissue
Cystic Duct
Fat
Gallbladder
Gastrointestinal Tract
Grasper
Hepatic Vein
L-hook Electrocautery
Liver
Liver Ligament
Class


Pixel Distribution Across Splits
Split
test
train
validation
Pixels (%)
0
5
10
15
20
25
30
Abdominal Wall
Background
Blood
Connective Tissue
Cystic Duct
Fat
Gallbladder
Gastrointestinal Tract
Grasper
Hepatic Vein
L-hook Electrocautery
Liver
Liver Ligament
Class

4. Qualitative Analysis — train — Sample 5906

Reference Image



Ground Truth

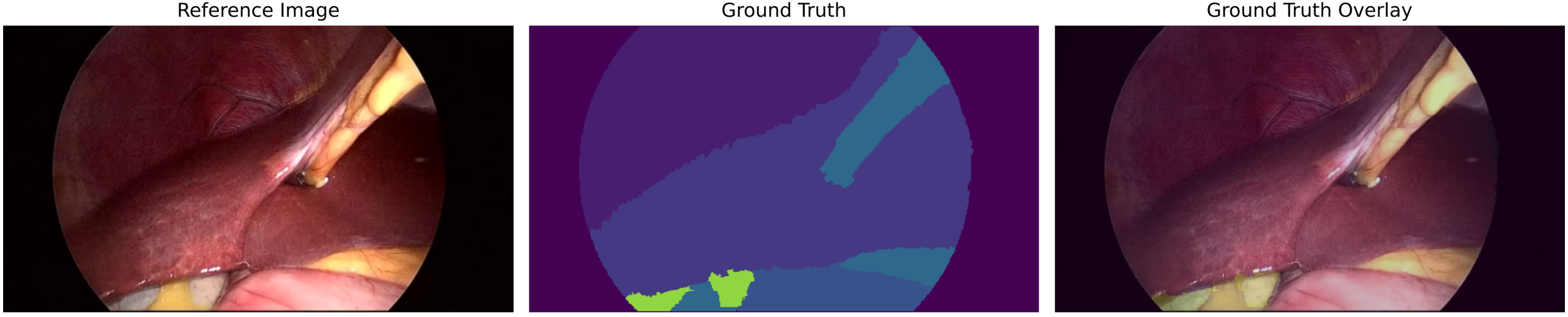


Ground Truth Overlay



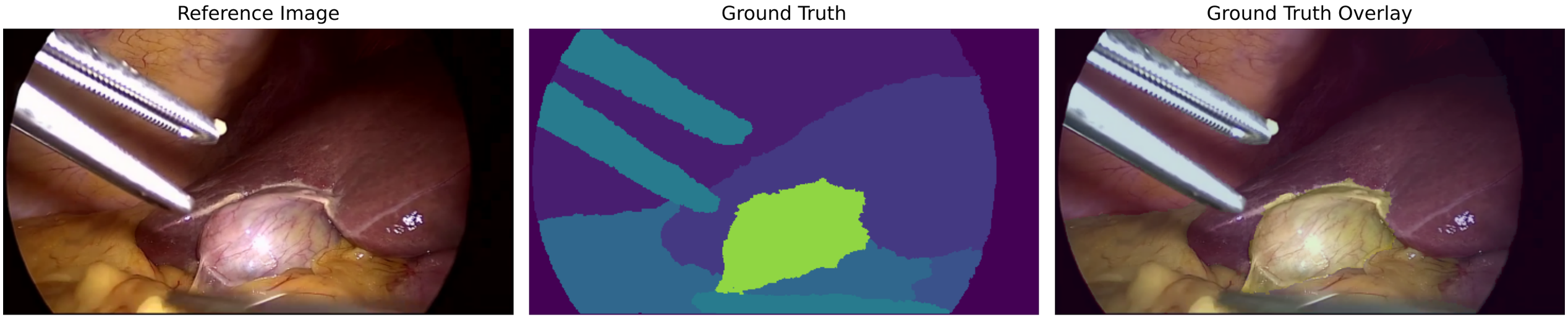
Present classes: Background, Abdominal Wall, Liver, Fat, Grasper, Gallbladder

4. Qualitative Analysis — train — Sample 4111



Present classes: Background, Abdominal Wall, Liver, Gastrointestinal Tract, Fat, Gallbladder

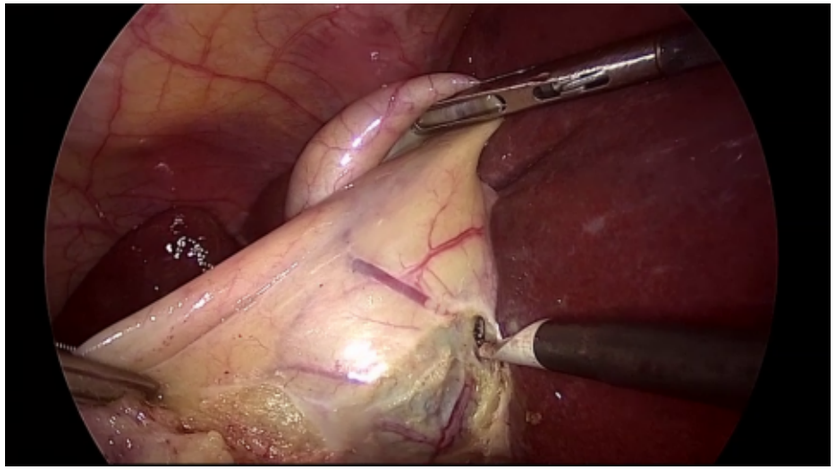
4. Qualitative Analysis — train — Sample 5267



Present classes: Background, Abdominal Wall, Liver, Gastrointestinal Tract, Fat, Grasper, Gallbladder

4. Qualitative Analysis — train — Sample 478

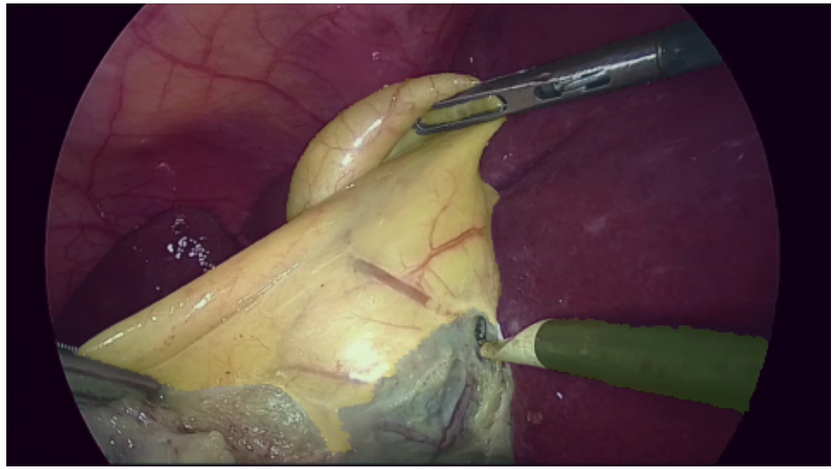
Reference Image



Ground Truth

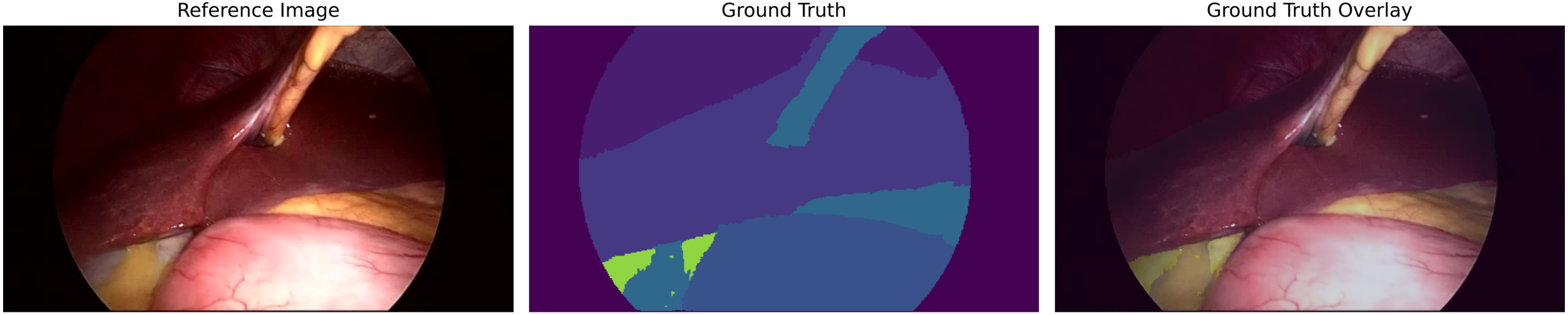


Ground Truth Overlay



Present classes: Background, Abdominal Wall, Liver, Grasper, Connective Tissue, L-hook Electrocautery, Gallbladder

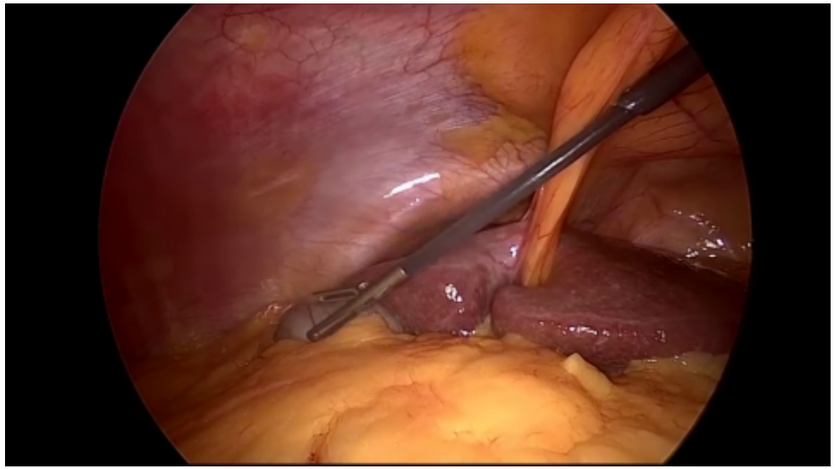
4. Qualitative Analysis — train — Sample 4080



Present classes: Background, Abdominal Wall, Liver, Gastrointestinal Tract, Fat, Gallbladder

4. Qualitative Analysis — validation — Sample 6422

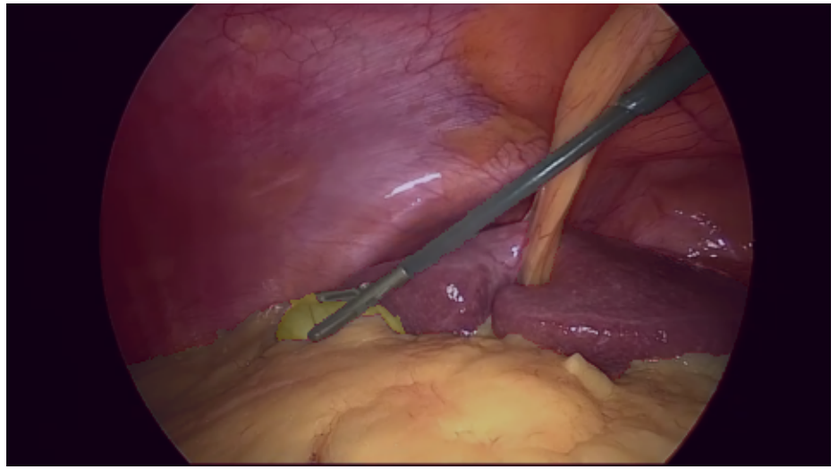
Reference Image



Ground Truth

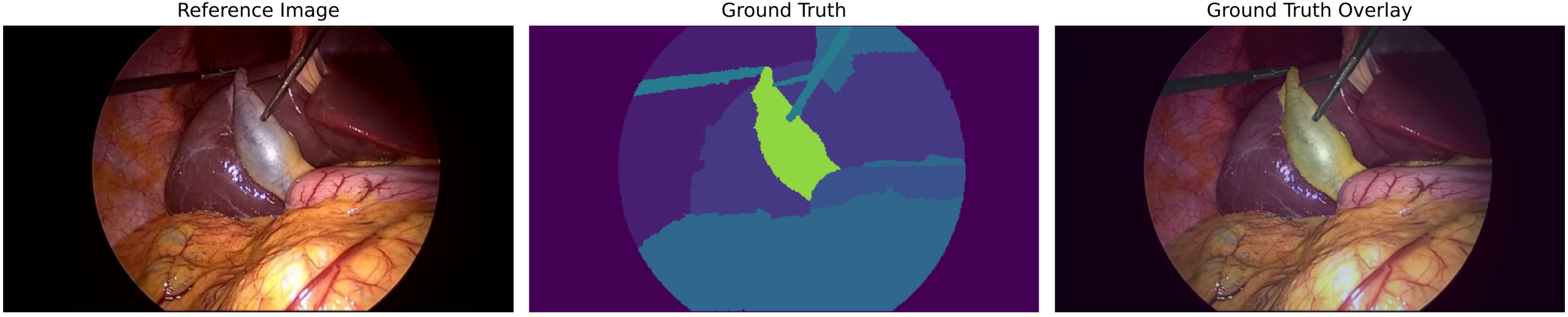


Ground Truth Overlay



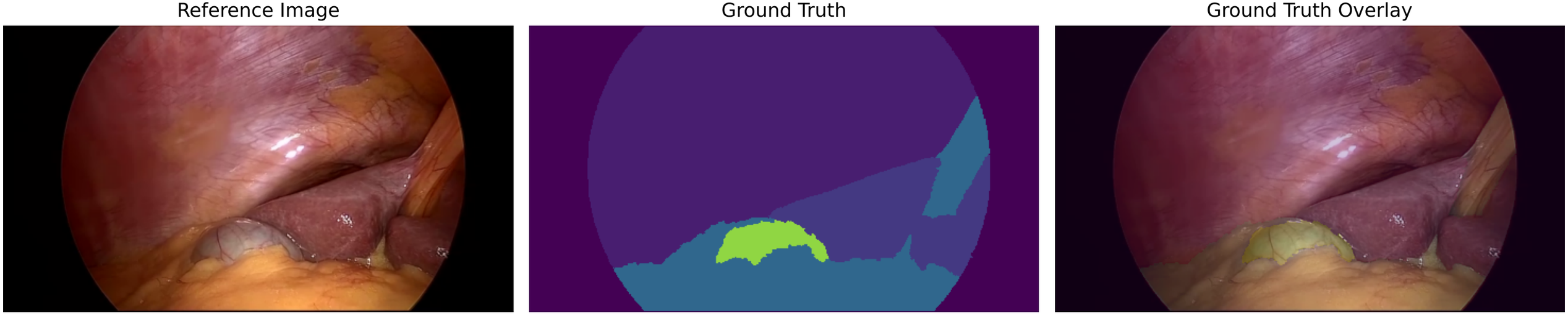
Present classes: Background, Abdominal Wall, Liver, Fat, Grasper, Gallbladder

4. Qualitative Analysis — validation — Sample 3513



Present classes: Background, Abdominal Wall, Liver, Gastrointestinal Tract, Fat, Grasper, Gallbladder

4. Qualitative Analysis — validation — Sample 6099



Present classes: Background, Abdominal Wall, Liver, Fat, Gallbladder

4. Qualitative Analysis — validation — Sample 2402

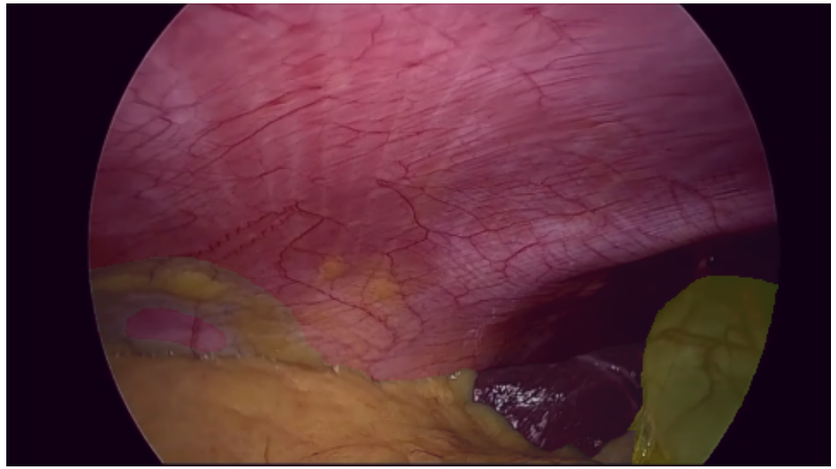
Reference Image



Ground Truth



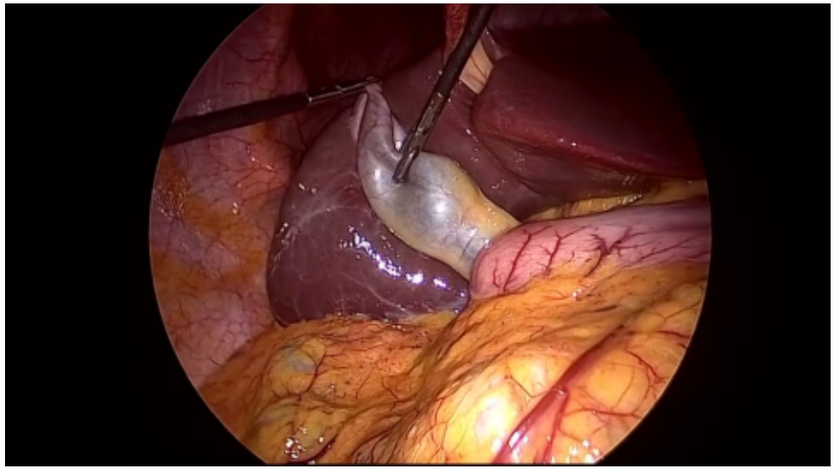
Ground Truth Overlay



Present classes: Background, Abdominal Wall, Liver, Fat, Gallbladder

4. Qualitative Analysis — validation — Sample 3497

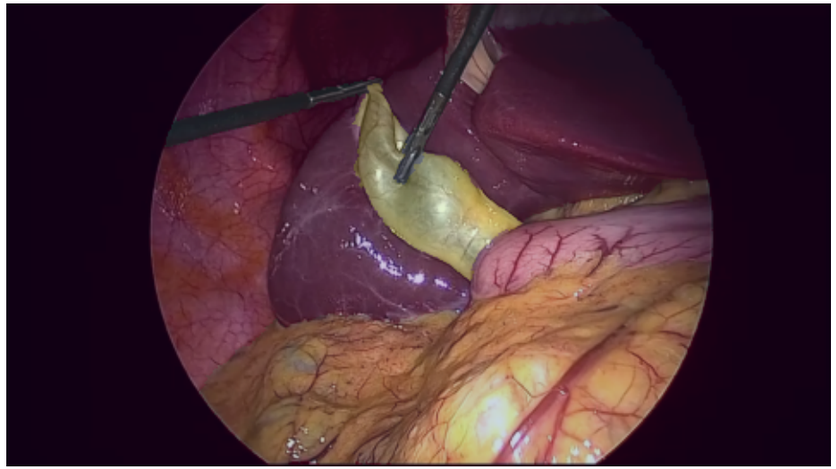
Reference Image



Ground Truth



Ground Truth Overlay



Present classes: Background, Abdominal Wall, Liver, Gastrointestinal Tract, Fat, Grasper, Gallbladder

5. Validation Summary

Dataset: cholec_seg8k

Number of classes: 13

Split sizes:

- train: 5,360 samples
- validation: 2,720 samples
- test: 0 samples

Class presence:

- train: Liver Ligament has the lowest sample presence (4.48%).
- validation: Connective Tissue has the lowest sample presence (0.00%).
- test: Background has the lowest sample presence (0.00%).

Pixel distribution:

- train: Background is the dominant class (24.39%), while Hepatic Vein has the smallest pixel share (0.0143%).
- validation: Background is the dominant class (32.87%), while Connective Tissue has the smallest pixel share (0.0000%).
- test: Background is the dominant class (0.00%), while Background has the smallest pixel share (0.0000%).

Interpretation:

- Sample presence measures how frequently a class occurs across images.
- Pixel distribution measures how much image area is occupied by each class.
- These two statistics describe different forms of class imbalance.
- Differences between splits should be considered when interpreting model performance.
- Group-level stratification cannot guarantee that every rare class appears in every split.